

ATF Field Specification v1.2

OMNIX QUANTUM LTD · Harold Alberto Nunes Rodelo · Day 1 Deliverable for VeriSigil AI · 22 May 2026

ATF FIELD SPECIFICATION Agent Trust Fabric — Trace Format & Field Definitions Version: 1.2 ·
Date: 2026-05-22 Issued by: Harold Alberto Nunes Rodelo, ATF / OMNIX QUANTUM LTD For:
VeriSigil AI (Raheem Larry Babatunde) — Bridge Validation Collaboration Status: Day 1
Deliverable — Sandbox / Evaluation Only

Overview

This document defines the complete field specification for the three core ATF record types used in the TAR↔TAP and RCR↔Survivability bridge validation:

- DR — Delegation Receipt (ATFDR-`{16HEX}`)
- TAR — Temporal Admissibility Record (ATFTAR-`{16HEX}`)
- RCR — Runtime Continuity Record (ATFRCR-`{16HEX}`)

All records are PQC-signed (Dilithium-3 / ML-DSA-65, FIPS 204). All `content_hash` values are SHA-256 hex digests (no prefix). All timestamps are ISO 8601 UTC unless noted as nanosecond epoch.

1. Delegation Receipt (DR)

Record ID format: `ATFDR-{16 uppercase hex characters}` ADR reference: ADR-156

1.1 Fields

Field	Type	Required	Description
<code>delegation_id</code>	string	REQUIRED	Unique DR identifier. Format: <code>ATFDR-<code>{16HEX}</code></code>
<code>delegator_id</code>	string	REQUIRED	AID or human operator ID issuing the delegation
<code>delegate_id</code>	string	REQUIRED	AID of the receiving (delegated) agent
<code>task_scope</code>	object	REQUIRED	Dict describing what the delegate is authorized to do. Arbitrary key-value pairs.
<code>authority_budget_delegator</code>	float	REQUIRED	Authority budget held by delegator at issuance. Range: 0.0-100.0
<code>authority_budget_granted</code>	float	REQUIRED	Authority budget granted to delegate. MUST be $\leq$ <code>authority_budget_delegator</code> (ATF-INV-001 / MAR).
<code>parent_delegation_id</code>	string	OPTIONAL	DR ID of the delegation that gave delegator its authority. Null if delegator is human root.

Field	Type	Required	Description
chain_root_id	string	REQUIRED	DR ID of the originating human-root delegation in the chain.
delegation_depth	integer	REQUIRED	Chain depth. 0 = human root, 1 = first agent, N = Nth sub-agent.
delegator_public_key	string	REQUIRED	Dilithium-3 public key of delegator, base64-encoded. Used for offline signature verification.
content_hash	string	REQUIRED	SHA-256 hex digest of all DR fields except content_hash, pqc_signature, pqc_algorithm. Computed over canonical JSON (keys sorted, no whitespace).
posture_state_hash	string	REQUIRED	SHA-256 of authority posture at exact moment of DR issuance. Committed fields: delegator_id, task_scope, authority_budget_delegator, authority_budget_granted, delegation_depth, chain_root_id, created_at. Enables SPV binding for TAR↔TAP bridge validation.
pqc_signature	string	CONDITIONAL	Dilithium-3 (ML-DSA-65) signature over content_hash, base64-encoded. Present when signing key is available.
pqc_algorithm	string	CONDITIONAL	Signing algorithm identifier. Value: "dilithium3"
expires_at	string	OPTIONAL	ISO 8601 UTC expiry timestamp. Null = no expiry.
status	string	REQUIRED	ACTIVE EXPIRED REVOKED
created_at	string	REQUIRED	ISO 8601 UTC timestamp of DR issuance.
metadata	object	REQUIRED	Extension dict. Empty object {} if unused.

1.2 Invariants

- ATF-INV-001 (MAR): `authority_budget_granted ≤ authority_budget_delegator`. Violation raises hard error before issuance — no DR is created.
- ATF-INV-002: `chain_root_id` traces back to a human-root delegation (`delegation_depth = 0`).
- ATF-INV-003: `posture_state_hash` is computed before `content_hash`. `content_hash` commits to `posture_state_hash`.

1.3 Example

```
{
  "delegation_id": "ATFDR-A1B2C3D4E5F60001",
  "delegator_id": "AID-COMPLIANCE-0001",
  "delegate_id": "AID-AUDIT-0007",
  "task_scope": {
```

```
    "domain": "compliance",
    "permitted_actions": ["read_evidence", "issue_tar"],
    "max_duration_s": 3600
  },
  "authority_budget_delegator": 80.0,
  "authority_budget_granted": 60.0,
  "parent_delegation_id": "ATFDR-R00T0000000001",
  "chain_root_id": "ATFDR-R00T0000000001",
  "delegation_depth": 1,
  "delegator_public_key": "<base64-dilithium3-pubkey>",
  "content_hash": "a3f1c2d4e5b6a7f8c9d0e1f2a3b4c5d6e7f8a9b0c1d2e3f4a5b6c7d8e9f0a1b2",
  "posture_state_hash": "b4e2d1c3f5a6b7c8d9e0f1a2b3c4d5e6f7a8b9c0d1e2f3a4b5c6d7e8f9a0b1c2",
  "pqc_signature": "<base64-dilithium3-signature>",
  "pqc_algorithm": "dilithium3",
  "expires_at": "2026-05-22T12:00:00+00:00",
  "status": "ACTIVE",
  "created_at": "2026-05-21T10:00:00+00:00",
  "metadata": {}
}
```

2. Temporal Admissibility Record (TAR)

Record ID format: `ATFTAR-{16 uppercase hex characters}` ADR reference: ADR-157

2.1 Fields

Field	Type	Required	Description
<code>tar_id</code>	string	REQUIRED	Unique TAR identifier. Format: <code>ATFTAR-{16HEX}</code>
<code>delegation_id</code>	string	REQUIRED	<code>ATFDR-{16HEX}</code> of the authorizing Delegation Receipt
<code>agent_id</code>	string	REQUIRED	AID of the acting agent. Format: <code>AID-{DOMAIN}-{16HEX}</code>
<code>execution_ref</code>	string	OPTIONAL	Reference to Execution Receipt (ADR-131). Null if not linked.
<code>execution_ns</code>	integer	REQUIRED	Nanosecond Unix timestamp of the execution event being admitted.
<code>execution_ts</code>	string	REQUIRED	ISO 8601 UTC timestamp of the execution event.
<code>dr_status_at_admission</code>	string	REQUIRED	Status of the DR at the moment of TAR admission. Value: <code>ACTIVE</code> <code>EXPIRED</code> <code>REVOKED</code>
<code>dr_expires_at</code>	string	OPTIONAL	DR expiry timestamp, copied from the DR.
<code>authority_budget</code>	float	REQUIRED	Authority budget of the DR at admission time.
<code>domain</code>	string	REQUIRED	Governance domain identifier (e.g. <code>"compliance"</code> , <code>"trading"</code> , <code>"medical_ai"</code>).
<code>task_action</code>	string	REQUIRED	

Field	Type	Required	Description
			The specific action being admitted (e.g. "read_evidence", "issue_receipt").
admission_status	string	REQUIRED	ADMITTED REJECTED
rejection_reason	string	OPTIONAL	Human-readable reason for rejection. Null if admitted.
content_hash	string	REQUIRED	SHA-256 hex digest of all TAR fields except content_hash, pqc_signature, pqc_algorithm.
pqc_signature	string	CONDITIONAL	Dilithium-3 (ML-DSA-65) signature over content_hash, base64-encoded.
pqc_algorithm	string	CONDITIONAL	"dilithium3"
chain_root_id	string	REQUIRED	chain_root_id of the authorizing DR. Enables chain-level traceability.
issued_at	string	REQUIRED	ISO 8601 UTC timestamp of TAR issuance.
metadata	object	REQUIRED	Extension dict.

2.2 Invariants

- ATF-INV-004: A TAR with admission_status = ADMITTED MUST reference a DR with dr_status_at_admission = ACTIVE.
- ATF-INV-005: execution_ns MUST be ≤ the nanosecond timestamp of issued_at.
- TAR is the primary bridge target for TAR↔TAP equivalence mapping. delegation_id is the join key.

2.3 Example

```
{
  "tar_id": "ATFTAR-F1E2D3C4B5A60001",
  "delegation_id": "ATFDR-A1B2C3D4E5F60001",
  "agent_id": "AID-COMPLIANCE-F1E2D3C4B5A60001",
  "execution_ref": null,
  "execution_ns": 1748131200000000000,
  "execution_ts": "2026-05-21T10:00:00+00:00",
  "dr_status_at_admission": "ACTIVE",
  "dr_expires_at": "2026-05-22T12:00:00+00:00",
  "authority_budget": 60.0,
  "domain": "compliance",
  "task_action": "read_evidence",
  "admission_status": "ADMITTED",
  "rejection_reason": null,
  "content_hash": "c5d6e7f8a9b0c1d2e3f4a5b6c7d8e9f0a1b2c3d4e5f6a7b8c9d0e1f2a3b4c5d6",
  "pqc_signature": "<base64-dilithium3-signature>",
  "pqc_algorithm": "dilithium3",
  "chain_root_id": "ATFDR-R00T0000000001",
  "issued_at": "2026-05-21T10:00:01+00:00",
  "metadata": {}
}
```

3. Runtime Continuity Record (RCR)

Record ID format: `ATFRCR-{16 uppercase hex characters}` ADR reference: ADR-159

3.1 Fields

Field	Type	Required	Description
<code>rcr_id</code>	string	REQUIRED	Unique RCR identifier. Format: <code>ATFRCR-{16HEX}</code>
<code>tar_id</code>	string	REQUIRED	<code>ATFTAR-{16HEX}</code> of the admission record this RCR monitors.
<code>delegation_id</code>	string	REQUIRED	<code>ATFDR-{16HEX}</code> of the authorizing DR.
<code>chain_root_id</code>	string	REQUIRED	Chain root delegation ID.
<code>agent_id</code>	string	REQUIRED	AID of the running agent.
<code>ces_score</code>	float	REQUIRED	Continuity Eligibility Score. Range: 0.0-100.0. Formula: $(T \times 0.30) + (B \times 0.30) + (D \times 0.20) + (I \times 0.20)$
<code>continuity_status</code>	string	REQUIRED	<code>NOMINAL</code> (75-100) <code>MONITORING</code> (50-75) <code>WARNING</code> (25-50) <code>CRITICAL</code> (10-25) <code>HALT</code> (0-10)
<code>ces_temporal</code>	float	REQUIRED	T component: $(\text{time remaining on DR} / \text{total DR lifetime}) \times 100$. Range: 0.0-100.0
<code>ces_budget</code>	float	REQUIRED	B component: $(\text{budget remaining} / \text{budget at admission}) \times 100$. Range: 0.0-100.0
<code>ces_context</code>	float	REQUIRED	D component: $100 - \text{context_drift_pct}$. Range: 0.0-100.0
<code>ces_integrity</code>	float	REQUIRED	I component: $100 - (\text{active_anomalies} \times 10)$. Range: 0.0-100.0
<code>time_remaining_ns</code>	integer	REQUIRED	Remaining DR lifetime in nanoseconds at this snapshot.
<code>dr_expires_at</code>	string	OPTIONAL	ISO 8601 UTC expiry timestamp of the DR. Null if DR has no expiry.
<code>fragmentation_score</code>	float	REQUIRED	Aggregate budget consumption across the delegation chain. Range: 0.0-1.0
<code>reauth_challenge_id</code>	string	OPTIONAL	Reference to ReauthorizationChallenge if issued (<code>ATFRC-{16HEX}</code>). Null otherwise.
<code>execution_ns</code>	integer	REQUIRED	Nanosecond Unix timestamp of the execution event.
<code>execution_ts</code>	string	REQUIRED	ISO 8601 UTC timestamp of the execution event.
<code>issued_at</code>	string	REQUIRED	ISO 8601 UTC timestamp of RCR issuance.
<code>budget_at_admission</code>	float	REQUIRED	Authority budget at the moment of TAR admission.
<code>budget_remaining</code>	float	REQUIRED	Authority budget remaining at this RCR snapshot.

Field	Type	Required	Description
context_drift_pct	float	REQUIRED	Context drift percentage at this snapshot. Range: 0.0–100.0
active_anomalies	integer	REQUIRED	Count of active governance anomalies at this snapshot.
content_hash	string	REQUIRED	SHA-256 hex digest of all RCR fields except content_hash , pqc_signature , pqc_algorithm .
pqc_signature	string	CONDITIONAL	Dilithium-3 (ML-DSA-65) signature over content_hash , base64-encoded.
pqc_algorithm	string	CONDITIONAL	"dilithium3"
sample_reason	string	OPTIONAL	Reason for this RCR snapshot: SCHEDULED EXECUTION_COMPLETE ANOMALY_DETECTED MANUAL .
escalation_event_id	string	OPTIONAL	Reference to ContinuityEscalationEvent if escalation was triggered.
predecessor_rcr_id	string	OPTIONAL	ID of previous RCR in the continuity chain.
metadata	object	REQUIRED	Extension dict.

3.2 Invariants

- RGC-INV-001: Every RCR MUST be anchored to a valid TAR (tar_id must resolve).
- RGC-INV-002: CES MUST be computed from real-time values only — no cached inputs.
- RGC-INV-003: HALT tier terminates execution and revokes in-flight sub-tasks.
- RGC-INV-005: All RCRs are PQC-signed and immutable after issuance.
- RGC-INV-006: RCR chain is acyclic — issued_at MUST be strictly monotonically increasing per execution session.
- RCR is the bridge target for RCR↔Survivability receipt mapping. Join key: tar_id .

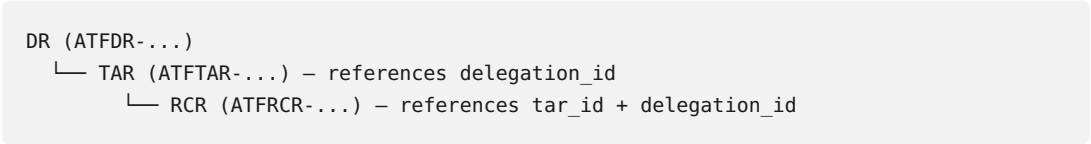
3.3 Example

```
{
  "rcr_id": "ATFRCR-D1C2B3A4E5F60001",
  "tar_id": "ATFTAR-F1E2D3C4B5A60001",
  "delegation_id": "ATFDR-A1B2C3D4E5F60001",
  "chain_root_id": "ATFDR-ROOT0000000001",
  "agent_id": "AID-COMPLIANCE-F1E2D3C4B5A60001",
  "ces_score": 92.7,
  "continuity_status": "NOMINAL",
  "ces_temporal": 87.0,
  "ces_budget": 92.0,
  "ces_context": 95.0,
  "ces_integrity": 100.0,
  "execution_ns": 1748131200000000000,
  "execution_ts": "2026-05-21T10:00:00+00:00",
  "issued_at": "2026-05-21T10:01:00+00:00",
  "budget_at_admission": 60.0,
  "budget_remaining": 55.2,
  "context_drift_pct": 5.0,
  "active_anomalies": 0,
  "time_remaining_ns": 9360000000000,
}
```

```
"dr_expires_at": "2026-05-22T12:00:00+00:00",
"fragmentation_score": 0.08,
"reauth_challenge_id": null,
"content_hash": "d6e7f8a9b0c1d2e3f4a5b6c7d8e9f0a1b2c3d4e5f6a7b8c9d0e1f2a3b4c5d6e7",
"pqc_signature": "<base64-dilithium3-signature>",
"pqc_algorithm": "dilithium3",
"sample_reason": "SCHEDULED",
"escalation_event_id": null,
"predecessor_rcr_id": null,
"metadata": {}
}
```

4. Cross-Record Traceability

The three records form a traceable chain:



Join path for TAR↔TAP bridge: TAR.delegation_id → DR.delegation_id → DR.posture_state_hash

Join path for RCR↔Survivability bridge: RCR.tar_id → TAR.tar_id → TAR.delegation_id → DR.chain_root_id

5. Signature Verification

All three record types use the same verification procedure:

- 1. Recompute content_hash from all fields except content_hash, pqc_signature, pqc_algorithm
- 2. Canonical JSON: json.dumps(fields, sort_keys=True, separators=(",", ":"))
- 3. SHA-256 hex of UTF-8 encoded canonical JSON
- 4. If pqc_signature is present: verify Dilithium-3 signature over content_hash.encode("utf-8") using the OMNIX platform public key

OMNIX platform public key (ML-DSA-65 / Dilithium-3, base64): Shared separately. See integration comms for current value.

6. Domain Values (Common)

Recognized domain values:

Value	Description
compliance	ISO 27001 / regulatory compliance
trading	Autonomous trading governance
medical_ai	Medical AI recommendation governance

Value	Description
defense_governance	Defense / national security AI
insurance	Insurance underwriting AI
real_estate	Real estate AI decisions
energy_governance	Energy sector AI
islamic_credit	Islamic finance credit AI
stablecoin	Stablecoin / DeFi governance
crisis	Crisis response AI
autonomous_agent	General autonomous agent

--

7. Revision History

Version	Date	Change
1.0	2026-05-21	Initial release — Day 1 deliverable for VeriSigil bridge validation
1.1	2026-05-22	RCR field names corrected to match implementation: ces→ces_score, ces_tier→continuity_status, temporal_health→ces_temporal, budget_health→ces_budget, context_fidelity→ces_context, integrity_score→ces_integrity. Added budget_at_admission, budget_remaining, context_drift_pct, active_anomalies, sample_reason, escalation_event_id, predecessor_rcr_id.
1.2	2026-05-22	CES component ranges corrected: 0.0-1.0 → 0.0-100.0 for ces_temporal, ces_budget, ces_context, ces_integrity. Component formulas corrected (no / 100 division). Added missing RCR fields: time_remaining_ns, dr_expires_at, fragmentation_score, reauth_challenge_id. Example JSON values corrected. Fixed RGC-INV-006 reference from issued_at_ns to issued_at.